POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Single electrode potential. How do you determine the single electrode potential of Ag – electrode experimentally? Explain. Calculate the emf of the cell at 27°C when given electrodes are coupled together. 2+3+3

$$Zn \rightarrow Zn^{++} + 2e^{\ominus} E^0 = +0.76V$$
$$Ag \rightarrow Ag^{+} + 2e^{\ominus} E^0 = -0.80V$$
$$R = 8.314JK^{-1}mol^{-1}, [Zn^{++}] = 0.2M, [Ag^{+}] = 0.1M$$

b) Define buffer solution. How does basic buffer resist the change in pH on adding few drops of acid and base? The pH of a buffer solution containing 0.5 mol/lit CH_3COOH and 0.5 mol/lit CH_3COONa is found to be 4.76. What will be the pH of this solution after adding 0.1 mol/lit HCl. Assume that the volume is unchanged.($Ka=1.75x10^{-5}$) 7

2. a) Give reasons for the followings: 8
   i. $TiCl_3$ is coloured but $TiCl_4$ is colourless.
   ii. Transition elements show variable oxidation state.
   iii. Zn^{+2} salts are white.
   iv. Cu is not considered as transition element.

b) Differentiate between Electron affinity and Electronegativity. Explain the factor affecting the Ionization potential. 7

OR

What are representative elements? Write their properties.

3. a) What are enantiomers and diasteriomers? Write the characteristics of enantiomers. What are free radicals? Explain the factors that stabilize the free radicals. 8

b) What are Electrophile and Nucleophiles? Write the product and mechanism of the following chemical reactions. 7
   i. $CH_3 - CH = CH_2 + HBr \xrightarrow{R\text{-}O\text{-}O\text{-}R}$
   ii. $(CH_3)_3C - Br + NaOH(aq) \longrightarrow$
   iii. $C_6H_6 + Conc.HNO_3 \xrightarrow{Conc.H_2SO_4}$

4. a) What is polymerization reaction? Write the different types of polymerization with suitable examples for each. 8

b) Write short notes on: 7
   i. PVC
   ii. Vulcanization of rubber

5. a) What is the principle behind mass spectroscopy? Show your acquaintance to parent peak and base peak. 8

b) What are lubricants? Explain different kinds of lubricants with example. 7

6. a) What are the causes of water pollution? Mention its effect on human health and also explain its controlling measures 8

b) Define air pollution. What are the factors responsible for ozone layer depletion and what are its hazardous effect? Give your opinion. 7

7. Write short notes on: **(Any two)** 2×5
   a) Corrosion
   b) Nitration
   c) Thermoplastics and Thermosetting plastics